Usefulness of brachial artery flow-mediated dilation to predict long-term cardiovascular events in subjects without heart disease.

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Endothelial dysfunction is considered an important prognostic factor in atherosclerosis. To determine the long-term association of brachial artery flow-mediated dilation (FMD) and adverse cardiovascular (CV) events in healthy subjects, we prospectively assessed brachial FMD in 618 consecutive healthy subjects with no apparent heart disease, 387 men (63%), and mean age 54 ± 11 years. After overnight fasting and discontinuation of all medications for ≥12 hours, FMD was assessed using high-resolution linear array ultrasound. Subjects were divided into 2 groups: FMD ≤11.3% (n = 309) and >11.3% (n = 309), where 11.3% is the median FMD, and were comparable regarding CV risk factors, lipoproteins, fasting glucose, C-reactive protein, concomitant medications, and Framingham 10-year risk score. In a mean clinical follow-up of 4.6 ± 1.8 years, the composite CV events (all-cause mortality, nonfatal myocardial infarction, hospitalization for heart failure or angina pectoris, stroke, coronary artery bypass grafting, and percutaneous coronary interventions) were significantly more common in subjects with FMD ≤11.3% rather than >11.3% (15.2% vs 1.2%, p = 0.0001, respectively). Univariate analysis demonstrated that the median FMD significantly predicted CV events (odds ratio 2.78, 95% CI 1.35 to 5.71, p <0.001). Multivariate analysis, controlling for traditional CV risk factors, demonstrated that median FMD was the best independent predictor of long-term CV adverse events (odds ratio 2.93, 95% CI 1.28 to 6.68, p <0.001). In conclusion, brachial artery median FMD independently predicts long-term adverse CV events in healthy subjects with no apparent heart disease in addition to those derived from traditional risk factor assessment.